



Gen AI and Agentic AI in Production

Week 3
Day 5

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

GitHub
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

Alex - the financial planner



Alex: The Agentic Learning Equities Explainer

What it is:

SaaS commercial application, financial planner

Done:

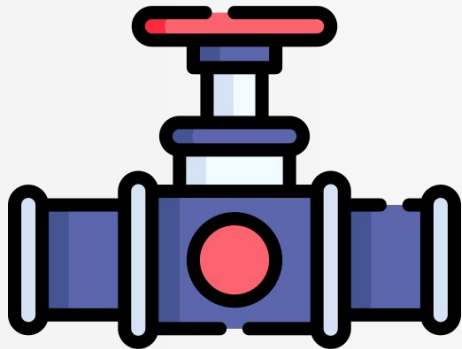
Sagemaker endpoint to convert text to vectors

Ingest lambda function to vectorize text and store in S3 Vectors

Todo:

Researcher Agent to carry out research and call ingest

Data Engineering



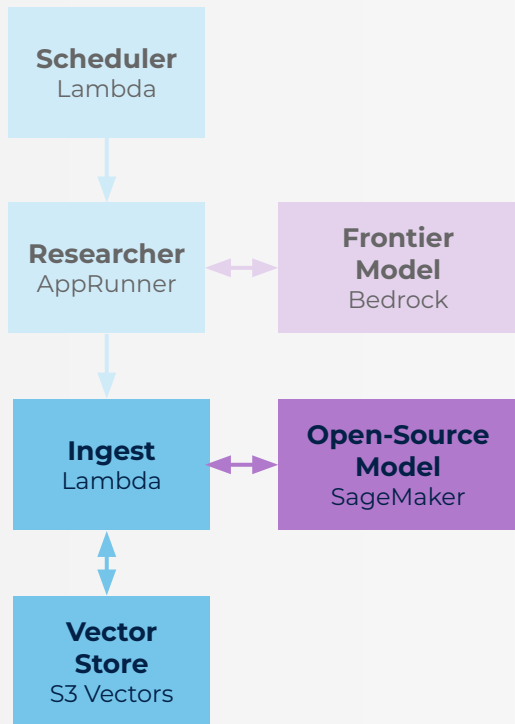
An entire discipline around building robust and scalable data pipes

Data pipelines & ETL (Extract, Transform, Load): Collect raw data, clean and transform it, and load it into storage where models can consume it

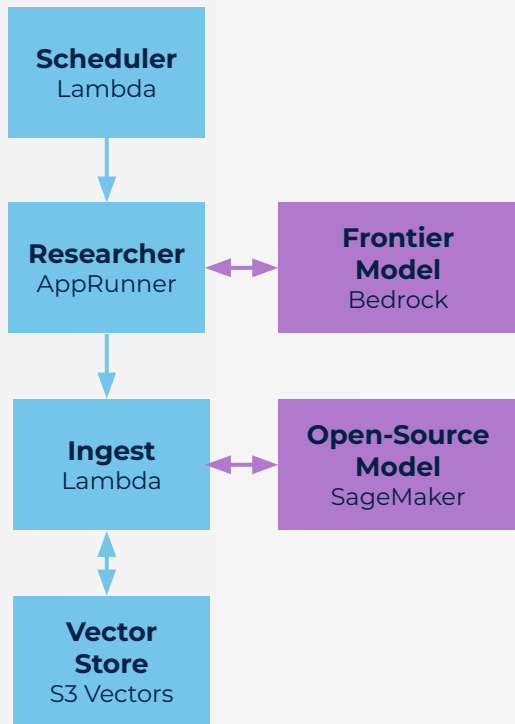
Scalable frameworks (e.g. Apache Spark, Apache Beam): Provide distributed computing power to process massive datasets efficiently, often in real time or batch modes.

Medallion architecture (Bronze → Silver → Gold): A common production pattern for enterprise-scale data transformation

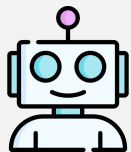
Data Ingest Architecture



Data Ingest Architecture



Assignment: one of:



Agentic AI

Add MCP servers (Polygon AI?); Context Engineering; Todo List



Platform Engineering

Investigate Amazon SQS for more robust task management



Data Engineering

Transform the data and store it in an external DB - eg Supabase

And then submit a PR with a notebook in **production/community_contributions** with a link to your repo

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75%
Complete